

2022-Ph-H1-Q18

Section: Particles and Waves

Topic: The Inverse Square Law

Radiation intensity is measured at various distances. Which statement is correct?

Answer: C

(Intensity is inversely proportional to distance squared)

Guidance for Students:

Intensity decreases with the square of distance: $I \propto \frac{1}{r^2}$. This applies to point sources like light or radiation.

Revision Tips:

- Inverse square law: double distance $\rightarrow$ quarter intensity.
- Applies to light, sound, gravity, and radiation.
- Graph of I vs $\frac{1}{r^2}$ is linear.